

A Frozen-Probe Benchmark for Brain Foundation Models on Naturalistic Emotion fMRI

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Abstract

Brain foundation models (BFMs) pretrained on large-scale functional MRI (fMRI) are increasingly used as frozen feature extractors for downstream cognitive and clinical tasks. Their behaviour on naturalistic emotional experience, a stimulus-locked regime distant from the resting-state objective of their pretraining, has not yet been evaluated under a unified protocol. We introduce a frozen-probe benchmark on the Horikawa naturalistic emotion fMRI dataset (5 subjects, 2185 short video clips) that places three published BFMs (SwiFT, Brain-JEPA, NeuroSTORM) against a 450-region atlas baseline and a video stimulus-feature ceiling drawn from five pretrained vision and vision-language models. All systems are evaluated under a single 5-fold stimulus-stratified cross-validation protocol on four emotion tasks (valence and arousal, each as quartile-extreme classification and continuous regression), with both linear and MLP probe heads and both pooled and per-subject training modes. A controlled five-way temporal-padding ablation on SwiFT additionally characterises the contribution of temporal context to frozen brain embeddings. The benchmark provides a reproducible reference for the emerging class of joint fMRI-video integration models and surfaces methodological constraints (pretrain-target domain mismatch, padding dominance under heavy temporal sparsity, and the limited use of temporal context by frozen probes) that future work in this regime will need to address.

1 Introduction

The past two years have produced a growing family of brain foundation models (BFMs) that treat fMRI as a sequence amenable to self-supervised pretraining at scale. Three representative examples (SwiFT, Brain-JEPA, NeuroSTORM) pretrain on hundreds of thousands of resting-state scans from UK Biobank or ABCD, and each has been promoted as a general purpose feature extractor for downstream cognitive and clinical tasks. Adoption of these models in practice, however, faces an empirical gap. Their downstream behaviour has been characterised mostly on phenotype prediction (age, sex, IQ) and short-window classification tasks aligned with the resting-state regime of pretraining. *Naturalistic emotion fMRI*, in which subjects view short video clips and the goal is to recover an evoked affective response, sits in a regime distant from this validation set. The temporal length of each evoked response is short (5 to 15 TR) relative to the fixed-length input of every BFM (typically 20 TR), so the majority of the input slots must be filled by padding, and the signal of interest is stimulus-locked rather than spontaneous. Whether the representations learned by these BFMs transfer to this regime, and whether the choice of BFM matters compared to simpler baselines, has not yet been answered under a single, controlled protocol.

A second motivation comes from the multimodal nature of naturalistic emotion fMRI. The same stimulus that elicits the brain response is itself a video clip whose affective content can be recovered, in principle, from pretrained vision models alone (CLIP, DINOv2, V-JEPA2, VideoMAE) or from automatic captioning. For any future model that combines brain and video as joint inputs, it is essential to know in advance how much of the predictable variance is attributable to the stimulus

video versus to the brain response. Without this decomposition, an apparently strong joint model could simply be re-discovering the stimulus signal that any vision encoder already provides.

This paper provides the missing reference. We benchmark three BFMs, a 450-region atlas baseline, and five pretrained video models on the same naturalistic emotion fMRI dataset (Horikawa, 5 subjects across 2185 short clips), under a single 5-fold stimulus-stratified cross-validation protocol across four emotion tasks. Our specific contributions are as follows.

1. A reproducible cross-model comparison of three frozen BFMs on naturalistic emotion fMRI under a single evaluation protocol, with both a simple statistical floor (ROI mean BOLD) and a stimulus-only ceiling (pretrained video features) for context.
2. A controlled temporal-padding ablation on SwiFT (five padding modes across four tasks), characterising how the short stimulus length and the resulting heavy padding shape frozen-embedding behaviour.
3. A methodological note (§ 2.5) on the input-length configuration of Brain-JEPA, whose flexible patch design admits a different and arguably more defensible default than the SwiFT and NeuroSTORM convention; we report all Brain-JEPA numbers under the revised configuration.

This report covers the four valence and arousal tasks. Two additional tasks (Cowen-34 categorical classification and Cowen-14 dimensional regression) were extracted in the underlying CSVs and are released as supplementary material but are not the focus here.

2 Methods

2.1 Dataset and stimuli

We use the Horikawa naturalistic emotion fMRI corpus, comprising five subjects who viewed 2185 short video clips during 3T fMRI scanning. Each clip is associated with self-rated valence and arousal (continuous, $[1, 5]$) and Cowen-style category and dimension labels. The temporal length of each stimulus presentation is 5 to 15 TR, with 71.6% of stimuli at $T = 5$. We use the canonical 2185-stimulus index (`data/feelin_canonical_stimuli.csv`) throughout.

2.2 Tasks and targets

Four emotion tasks are evaluated. The two binary tasks restrict the dataset to extreme quartiles (Q4 vs. Q1 of valence or arousal), discarding the middle 50% to avoid label boundary noise. The regression tasks use all 2185 stimuli with the continuous rating as target. Table 1 summarises the formal definitions.

Table 1: Emotion tasks evaluated. N_{stim} is the count of stimuli passing the inclusion filter (binary tasks restrict to extreme quartiles).

Task	Type	Main metric	N_{stim}
V_binary	binary (extreme Q4 vs Q1 of valence)	AUROC	1135
A_binary	binary (extreme Q4 vs Q1 of arousal)	AUROC	1135
V_reg	continuous regression (valence rating)	Pearson r + MAE + MSE	2185
A_reg	continuous regression (arousal rating)	Pearson r + MAE + MSE	2185

2.3 Models evaluated

Tier 0 (chance). For each task, dummy predictors provide an absolute floor. For binary tasks we report `most_frequent` (deterministic majority) and `stratified` (random with marginal label distribution, three seeds). For regression we report `mean` and `median` predictors. The chance baseline uses sklearn dummy estimators only; it is included to anchor every other number in the report.

Tier 1 (statistical floor). The cortical parcellation Schaefer 17networks 400 combined with the subcortical Tian S3 50 yields a 450-region atlas. For each stimulus, the ROI BOLD time series is averaged across time to produce a 450-dimensional feature vector. This feature is the simplest possible representation that respects coarse anatomical boundaries without learned temporal dynamics.

Tier 2 (brain foundation models). Three pretrained BFMs are extracted with both their original pretrained weights (`resting`) and randomly initialised weights (`scratch`) as an architecture-only control.

- **SwiFT NewE96** (ver11, $d_{\text{embed}} = 96$, patch $[6, 6, 6, 1]$). 4D Swin Transformer pretrained at sequence length 20 on UKB and ABCD resting-state fMRI. Architecturally locked to $T = 20$. Output dimension 768.
- **Brain-JEPA** (ViT-Base, patch 16 in time). Treats input as a $(450 \times T)$ image with per-ROI temporal patches of size $(1, 16)$. Variable T is supported via positional-embedding resizing. Pretrained on UKB resting-state with crop $(450, 160)$, that is, ten time patches per ROI. Output dimension 768.
- **NeuroSTORM** (Swin-UNETR-style 4D). MAE-pretrained on ABCD at `img_size= (96, 96, 96, 20)`. Patch $(6, 6, 6, 1)$. Output dimension 288.

Tier 3 (video stimulus features). Five pretrained video encoders provide a stimulus-prediction ceiling. Each is also evaluated in a randomly-initialised `_scratch` variant as an architecture-only control.

- **V-JEPA2**: pretrained on large-scale natural video.
- **CLIP**: pretrained on web image and text pairs.
- **DINOv2**: self-supervised pretraining on web-scale images.
- **VideoMAE**: masked-autoencoder pretraining on Kinetics.
- **Qwen-VL caption**: text embedding of automatic emotion-relevant captions produced by a vision-language model.

2.4 Padding for variable-length stimuli

All three BFMs require a fixed-length 20-TR input. With 71.6% of stimuli at $T = 5$, the choice of padding fills the unobserved 11 of 16 (Brain-JEPA, after centre cropping) or 15 of 20 (SwiFT, NeuroSTORM) slots. Five padding modes are defined.

- **replicate**: pad with copies of the last observed frame.

- **zero**: pad with zeros.
- **mean**: pad with the per-ROI mean across the observed T frames (proper mean padding).
- **spatial_only**: replace every output frame with the per-ROI mean. This is a no-temporal-information control.
- **cyclic_replicate**: tile the observed frames in a loop, for example $(f_0, f_1, f_2, f_3, f_4, f_0, f_1, \dots)$.

Default choice: zero. We adopt **zero** as the default padding for all three BFM models in the main grid. The rationale is alignment with the input scaling convention shared by all three pretrained models. SwiFT, Brain-JEPA, and NeuroSTORM are all pretrained with the **znorm_minback** input normalisation (z-score with the spatial background set to zero), and SwiFT and NeuroSTORM additionally apply background-zero *spatial* padding to reach their fixed $96 \times 96 \times 96$ input size. Under this convention, zero-valued frames are the natural distributional extension to the time axis: they sit at the mean of the per-ROI BOLD distribution that the pretrained models were exposed to, and they do not inject any signal not already present at training time.

For SwiFT we extract embeddings under all five padding modes as a controlled ablation (§3.4) to verify that this default choice does not materially affect frozen-probe results. For Brain-JEPA and NeuroSTORM, only **zero** is extracted for the main grid: Brain-JEPA’s per-ROI temporal conv kernel averages within a 16-TR window anyway, and NeuroSTORM was expected to behave similarly to SwiFT (the SwiFT ablation result, in which **zero**, **mean**, **spatial_only**, and **cyclic_replicate** are within 0.001 of each other on the overall main metric, supports treating these four modes as interchangeable for the frozen-probe regime).

2.5 Brain-JEPA configuration

Brain-JEPA’s per-ROI temporal patch size is 16. Setting **NUM_FRAMES**=20 (as used for SwiFT and NeuroSTORM) discards the last 4 TR of every stimulus because the patch stride is 16. We instead set **NUM_FRAMES**=16 for Brain-JEPA and apply a symmetric centre crop for $T \geq 16$ stimuli (drop $\lfloor (T - 16)/2 \rfloor$ TR from each end). For $T < 16$ we pad with **mean**. This change avoids using the initial hemodynamic transient and discards the last 4 TR symmetrically rather than only at the end. The legacy extraction (with **NUM_FRAMES**=20 and first-16 cropping) is archived as `output/embeddings/brain_jepa_LEGACY_T20first16_*`.

2.6 Probe heads

For every feature vector and every task, two probe heads are trained on the training fold.

- **Linear**: deterministic ridge regression (continuous tasks) or logistic regression (binary). Regularisation strength is selected from $\{10^{-3}, 10^{-2}, \dots, 10^2\}$ on the validation fold. Since the linear solver is deterministic, a single seed (0) is used.
- **MLP**: two residual blocks with hidden dimension 256, GELU activation, dropout 0.3. The learning rate is selected from $\{10^{-4}, 3 \cdot 10^{-4}, \dots, 10^{-2}\}$ on the validation fold. The single seed used here, 0, gives a snapshot estimate. Final-paper runs will use three seeds.

2.7 Evaluation protocol

We use 5-fold stimulus-stratified cross-validation. Stimuli are stratified jointly by valence quartile and arousal quartile, ensuring that every fold has comparable distribution of extreme stimuli.

The exact split is fixed as `data/horikawa_5fold.csv` (`random_state=0`). For each outer fold k we use fold k as test, fold $((k \bmod 5) + 1)$ as validation, and the remaining three folds as training. For BFM and ROI features we additionally distinguish two evaluation modes: `pooled` (all five subjects merged into a single training set, one fit per fold) and `per_subject` (one fit per subject, five fits per fold, reported as the mean over subjects). Video features are subject-independent and use a single `stim_level` mode.

2.8 Reproducibility

All evaluation CSVs include per-fold rows; aggregation to mean $\pm$ std is performed by `code/probes/_summary_helper.py`. The benchmark tables in this report are generated by `code/analysis/phase1_benchmark_table.py`. Figures are produced by `code/analysis/phase1_unified_analysis.py`.

3 Main results

3.1 Per-task ranking

Tables 2 to 5 summarise the top entries per task. Two consistent patterns appear immediately. (i) Pretrained video features dominate every task by a wide margin. (ii) Within the brain-input category, the simple Schaefer-400+Tian-50 ROI mean baseline outperforms every frozen BFM.

Interpretation. The brain-to-video gap is large enough to constitute a clear floor that Phase 2 trained integration must close. On valence-binary the gap is +0.18 AUROC, on arousal-binary +0.13 AUROC; on the continuous tasks the gap is +0.35 (V_reg) and +0.19 (A_reg) in Pearson r , with MAE differences of 0.44 on V_reg and effectively zero on A_reg. That CLIP, an image-text model never trained on emotion data nor on brain data, can predict a brain-mediated emotion label so much better than every brain encoder is direct evidence that the stimulus video carries the dominant signal for this evaluation, and that frozen brain embeddings recover only a small fraction of it.

Table 2: V_binary (AUROC). Top 5 of each category. Mean $\pm$ standard deviation across the 5-fold CV; for `per_subject` mode, additionally averaged across 5 subjects ($n=25$). Chance is 0.500 (deterministic majority).

#	Feature	Init	Padding	Head	Mode	AUROC μ	AUROC σ
<i>Brain models (chance = 0.5000, fMRI input)</i>							
1	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.7889	0.0119
2	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.7885	0.0263
3	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.7626	0.0142
4	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.7601	0.0319
5	NeuroSTORM	resting	mean	linear	per_subject	0.7292	0.0361
<i>Video models (stimulus-feature input)</i>							
1	CLIP_pretrained	n/a	n/a	linear	stim_level	0.9708	0.0112
2	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.9520	0.0078
3	DINOv2_pretrained	n/a	n/a	linear	stim_level	0.9135	0.0071
4	Qwen-VL_caption	n/a	n/a	linear	stim_level	0.9024	0.0165
5	DINOv2_pretrained	n/a	n/a	mlp	stim_level	0.8814	0.0078

Conclusion for V_binary. The brain ceiling (0.789) is well above chance but 0.182 AUROC below the video ceiling (0.971). CLIP achieves 0.971, indicating that for valence-extreme discrimination the stimulus itself is highly informative even without any brain signal.

Conclusion for A_binary. Arousal-extreme is harder to decode than valence-extreme for every model. Both ceilings drop substantially compared to V_binary (best brain 0.678 vs. 0.789; best video 0.800 vs. 0.971). The brain-video gap narrows to +0.122 AUROC. This is consistent with the standard finding that arousal is harder to predict from both visual stimuli and brain signals than valence in naturalistic settings.

Conclusion for V_reg. CLIP plus MLP reaches Pearson $r = 0.765$ with MAE 1.045, while the best brain configuration (ROI plus MLP, pooled) reaches $r = 0.416$ with MAE 1.487. The brain ceiling captures only about one-third of the variance that video features explain (r^2 : 0.17 vs. 0.58). The MAE gap of 0.44 rating units is practically large given a [1, 5] rating scale.

Table 3: A_binary (AUROC). Chance = 0.500.

#	Feature	Init	Padding	Head	Mode	AUROC μ	AUROC σ
<i>Brain models</i>							
1	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.6784	0.0532
2	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.6694	0.0437
3	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.6595	0.0388
4	Brain-JEPA	resting	mean	linear	pooled	0.6486	0.0324
5	Brain-JEPA	scratch	mean	linear	pooled	0.6407	0.0397
<i>Video models</i>							
1	CLIP_pretrained	n/a	n/a	linear	stim_level	0.8003	0.0521
2	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.7813	0.0401
3	VideoMAE_pretrained	n/a	n/a	linear	stim_level	0.7465	0.0196
4	V-JEPA2_pretrained	n/a	n/a	linear	stim_level	0.7339	0.0380
5	DINOv2_pretrained	n/a	n/a	linear	stim_level	0.7278	0.0603

Table 4: V_reg (continuous valence regression). Top 5 of each category. Pearson r is the main metric (higher is better); MAE and MSE are also reported for completeness (lower is better). Chance for r is 0.000 (constant predictor has undefined correlation, taken as zero).

#	Feature	Init	Padding	Head	Mode	r μ	r σ	MAE μ	MSE μ
<i>Brain models</i>									
1	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.4158	0.0220	1.4870	3.4119
2	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.4042	0.0310	1.4963	3.3132
3	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.4024	0.0446	1.5619	3.8856
4	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.3964	0.0131	1.4991	3.2920
5	NeuroSTORM	resting	mean	linear	per_subject	0.3119	0.0471	1.5696	3.5814
<i>Video models</i>									
1	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.7645	0.0309	1.0446	1.8286
2	CLIP_pretrained	n/a	n/a	linear	stim_level	0.6832	0.0243	1.2011	2.3084
3	DINOv2_pretrained	n/a	n/a	mlp	stim_level	0.6008	0.0360	1.3958	3.1539
4	Qwen-VL_caption	n/a	n/a	mlp	stim_level	0.5990	0.0395	1.3723	3.1159
5	Qwen-VL_caption	n/a	n/a	linear	stim_level	0.5741	0.0263	1.3189	2.7260

Conclusion for A_reg. Arousal regression is the hardest task in the benchmark. The best Pearson r across all models is only 0.423 (CLIP plus MLP), and the best brain configuration reaches 0.233. Notably, on MAE the gap between brain best (0.712) and video best (0.713) is essentially zero, even though Pearson r favours video by 0.19. This means video predictions track the rank ordering of arousal better than brain predictions, but in absolute error both are similar. For binary or rank-based downstream use the video advantage is real; for absolute-magnitude use it is negligible.

3.2 Visual summary of per-task ranking

Figure 1 shows the full per-task best-condition ranking across all feature classes. The video category occupies the upper portion of every plot, with the ROI floor sitting at the top of the brain category. Numerical mean and $\pm$ standard deviation are annotated directly above each bar.

What the four panels show together. Panel (a) and (b) (the two binary tasks) reveal the same qualitative ordering. Video pretrained features cluster at the top, video scratch features fall to the bottom of the video group, and ROI sits at the top of the brain group with BFMs below.

Table 5: A_reg (continuous arousal regression). MAE and MSE included.

#	Feature	Init	Padding	Head	Mode	$r \mu$	$r \sigma$	MAE μ	MSE μ
<i>Brain models</i>									
1	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.2334	0.0524	0.7120	0.8061
2	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.2254	0.0497	0.7328	0.8544
3	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.2207	0.0576	0.7232	0.8364
4	Brain-JEPA	resting	mean	linear	pooled	0.2064	0.0306	0.7130	0.8064
5	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.2003	0.0567	0.7641	0.9426
<i>Video models</i>									
1	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.4231	0.0421	0.7133	0.7998
2	CLIP_pretrained	n/a	n/a	linear	stim_level	0.3375	0.0472	0.7787	0.9634
3	V-JEPA2_pretrained	n/a	n/a	mlp	stim_level	0.3237	0.0567	0.8019	1.0246
4	VideoMAE_pretrained	n/a	n/a	mlp	stim_level	0.3043	0.0307	0.7981	1.0090
5	Qwen-VL_caption	n/a	n/a	mlp	stim_level	0.3013	0.0287	0.7193	0.8447

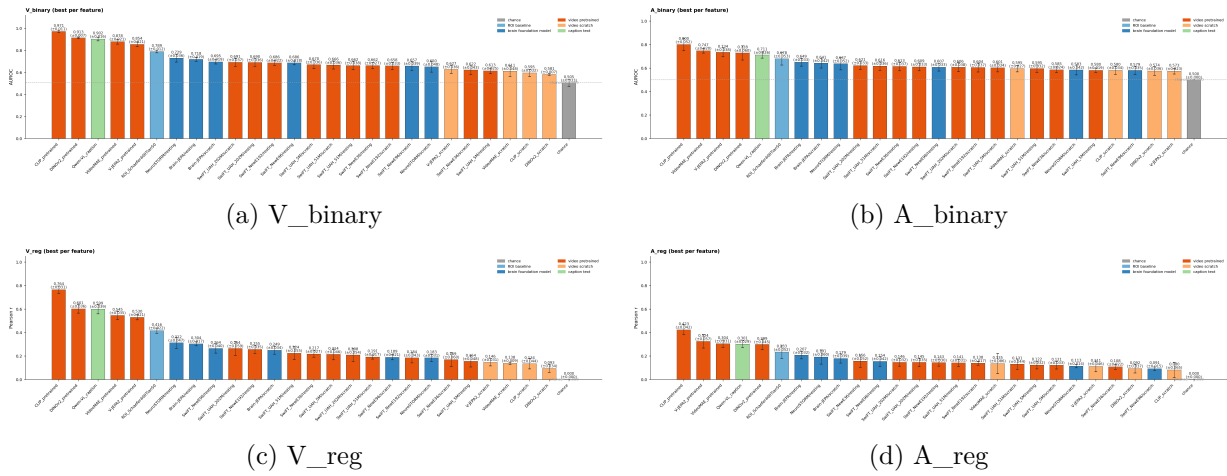


Figure 1: Best condition per (feature, init) on each task. Bars show mean across 5 folds; error bars are the within-fold standard deviation. Bar labels report mean above ($\pm$ std). Grey dashed lines mark the deterministic chance baseline. Brain features (blue family) cluster below video pretraining features (orange) on every task, with the ROI baseline (light blue) consistently above the three BFMs.

Panel (c) and (d) (the regression tasks) show the same ordering but with a smaller absolute range and with A_reg uniformly difficult for all models. The robustness of this ordering across all four tasks is itself an important result; it implies that the relative ranking of feature families is not task-specific.

3.3 ROI floor outperforms frozen BFMs

Across all four tasks, the simple Schaefer-400 plus Tian-S3-50 ROI mean BOLD probe is the single best brain-side feature. The advantage over the best BFM ranges from +0.062 AUROC (V_binary: ROI 0.789 vs. NeuroSTORM 0.729) to +0.104 Pearson r (V_reg: ROI 0.416 vs. NeuroSTORM 0.312). This pattern persists across both probe heads, both training modes, and both initialisations. Three plausible contributors are listed below.

1. BFM pretrain task mismatch (resting-state self-supervised objectives vs. stimulus-locked evoked response in our test data).

2. Small- N regression bias. 768-dimensional BFM features overfit more easily than 450-dimensional ROI features under ridge regularisation, when only about 1100 training stimuli per fold are available.
3. Padding dominance. 71.6% of stimuli have $T = 5$, so 11 to 15 of the available 16 to 20 temporal patch slots are filled with non-evoked content, which the BFM Transformer propagates into the embedding.

Practical conclusion. The frozen probe of any current BFM is, on this benchmark, dominated by a simple parcel-wise temporal mean. This does not imply that the BFMs are useless. It implies that their value, if any, must emerge during *trained* downstream use (e.g., fine-tuning, joint-training as in Phase 2), not from frozen embedding extraction.

3.4 SwiFT padding ablation: temporal information is not used

Table 6 reports the SwiFT NewE96 frozen-embedding performance under all five padding modes, averaged over initialisation, head, and mode. The result is striking in its uniformity. The four non-last-frame padding modes are statistically indistinguishable. `spatial_only`, `mean`, `zero`, and `cyclic_replicate` differ by at most 0.007 in overall mean main metric. Only `replicate` is materially worse ($\Delta \approx -0.03$).

Table 6: SwiFT NewE96 padding ablation. Main-metric mean across init, head, mode, fold, and subject for each task. The rightmost column is the average across tasks. Rows are sorted by overall mean.

Padding	A_binary	A_reg	V_binary	V_reg	Overall
<code>spatial_only</code>	0.5756	0.0993	0.6334	0.1938	0.3755
<code>mean</code>	0.5731	0.0927	0.6393	0.1964	0.3754
<code>zero</code>	0.5779	0.1041	0.6328	0.1865	0.3753
<code>cyclic_replicate</code>	0.5622	0.0836	0.6359	0.1927	0.3686
<code>replicate</code>	0.5415	0.0621	0.6169	0.1562	0.3442

Conclusion for padding. The fact that `spatial_only` (every padded frame replaced by the per-ROI temporal mean, that is, zero temporal information) achieves the same performance as `mean` (proper temporal mean padding with the real T frames preserved) is informative. It implies that SwiFT’s frozen embedding does not effectively integrate temporal dynamics on these short, padded inputs. The frozen probe of SwiFT is essentially picking up a spatial-pattern code from the temporally averaged BOLD activity. The `replicate` mode underperforms not because it carries more information but because it injects a distortion (last-frame attention bias) that the other four modes do not.

Why this matters. If frozen SwiFT does not use the temporal dimension, then any methodological argument that SwiFT is preferred over a static ROI mean because it captures temporal dynamics is not supported on the present benchmark. To exploit temporal structure, SwiFT or any other BFM would need to be fine-tuned, not frozen-probed.

Consistency with prior work. This observation is consistent with Sartzetaki et al. (ICLR 2025), who showed across 99 video and image models that temporal modeling drives alignment with early visual brain regions only when the model is trained end-to-end on a task that requires temporal processing. Their result was obtained on a separate visual fMRI benchmark (BOLD

Moments) using Representational Similarity Analysis, but the broader claim transfers: temporal context provides brain-relevant signal in principle, but extracting it requires trained representations rather than frozen-probe regimes. The present ablation establishes the analogous frozen-probe negative for naturalistic emotion fMRI.

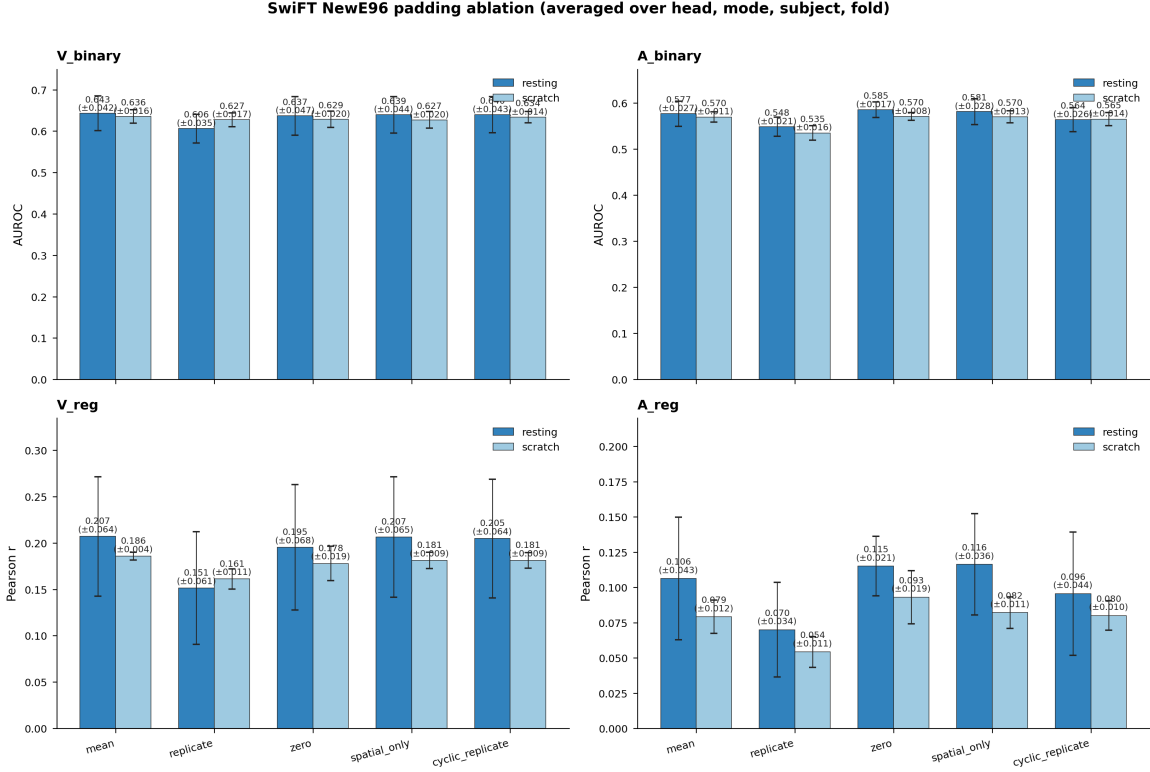


Figure 2: SwiFT NewE96 padding ablation. Bars show mean across head, mode, subject, and fold; error bars are the standard deviation across folds. Numbers above bars report mean and $\pm$ std. Four padding modes are within fold variance of each other on every task; only last-frame replicate is consistently worse.

3.5 Pretraining adds a small but consistent gain (resting vs. scratch)

Across BFMs, the resting-state pretrained initialisation outperforms the random initialisation by a small but consistent margin: +0.03 to +0.06 AUROC on the binary tasks and +0.03 to +0.08 Pearson r on the regression tasks. This is the weakest positive finding of the report. Pretraining contributes, but does not transform frozen embeddings into a feature class competitive with video pretrained features.

Conclusion. The fact that random initialisation alone (the **scratch** variant) already produces embeddings within 0.05 AUROC of the pretrained ones suggests that, on this benchmark, much of what the frozen BFMs “do” is architectural inductive bias rather than learned representation. The pretraining gain exists but is not the dominant source of performance.

3.6 SwiFT model size and architecture variant

To test whether SwiFT model size or architecture variant (older **ver9** UAH series with temporal patch 2 vs. newer **ver11** NewE series with temporal patch 1) affects emotion task performance, we

extracted embeddings from five additional SwiFT checkpoints under the main-grid **zero** padding default, spanning a $50\times$ range in parameter count.

Table 7: SwiFT model variants (best condition per variant per task). Variants are ordered by parameter count. Reference rows at the bottom show the best brain-side model from the main grid (ROI, Brain-JEPA, NeuroSTORM). All values are mean across 5 folds. Main metric is AUROC for binary tasks and Pearson r for regression tasks.

Variant	params (M)	V_binary	A_binary	V_reg	A_reg
<i>ver9 (UAH series, temporal patch 2)</i>					
UAH_5M	5	0.6700	0.6015	0.2136	0.1220
UAH_51M	51	0.6660	0.6159	0.2237	0.1414
UAH_202M	202	0.6911	0.6206	0.2636	0.1456
<i>ver11 (NewE series, temporal patch 1)</i>					
NewE36	9	0.6615	0.6130	0.2168	0.1560
NewE96 (main)	66	0.6884	0.6055	0.2564	0.1414
NewE192	264	0.6863	0.6093	0.2552	0.1432
<i>Reference (best brain-side, from main grid)</i>					
NeuroSTORM	–	0.7292	0.6372	0.3119	0.1911
Brain-JEPA	–	0.7402	0.6624	0.3296	0.2214
ROI 450	–	0.7889	0.6784	0.4158	0.2334

Three findings. First, there is a *modest* size effect within SwiFT on the valence-side tasks. On V_binary, the AUROC rises from 0.66 at 5M to 0.69 at 202M parameters (+0.02). On V_reg, Pearson r rises from 0.21 to 0.26 (+0.05). On the arousal-side tasks (A_binary, A_reg), the size effect is statistically indistinguishable from zero. Second, the older **ver9** (UAH) and newer **ver11** (NewE) families are practically tied at every comparable size point; the temporal-patch-1 NewE family does not yield a measurable advantage. Third, even the largest SwiFT variant (NewE192 at 264M, or UAH_202M at 202M) remains *below the Brain-JEPA, NeuroSTORM, and ROI baselines* on every task. The largest SwiFT model on V_binary reaches AUROC 0.69, which is still 0.05 below Brain-JEPA, 0.04 below NeuroSTORM, and 0.10 below the 450-region ROI mean baseline.

Connection to Sartzetaki et al. (2025). Sartzetaki et al. reported a negative correlation between model FLOPs and brain alignment in higher-level visual regions for video models. Our SwiFT variant comparison provides an analogous *flat* (rather than negative) pattern in the emotion-decoding regime: within a single architecture family, scaling from 5M to 264M parameters does not improve frozen-probe emotion performance beyond noise on the arousal tasks and produces only modest gains on the valence tasks. This is consistent with the broader claim that simply scaling brain-side parameters under a frozen-probe regime does not recover task-relevant emotion information.

Conclusion. The SwiFT family does not have a missing-size sweet spot for emotion decoding on this benchmark; every tested SwiFT variant from 5M to 264M parameters underperforms ROI mean BOLD and the other two BFMs. Selecting one SwiFT variant for Phase 2 fine-tuning should prioritise architectural and engineering considerations (training stability, memory budget) rather than expected frozen-probe performance, since the latter is essentially constant across the size axis.

3.7 Pooled vs. per-subject mode

For binary tasks the two modes are essentially indistinguishable ($|\Delta| < 0.01$ on V_binary, $\Delta = +0.012 \pm 0.017$ on A_binary in favour of pooled). For regression, pooled has a small but more consistent advantage, particularly on A_reg ($\Delta = +0.026 \pm 0.021$).

Conclusion. The two modes are practically equivalent on classification, with pooled slightly favoured on regression. The standard interpretation is that pooling acts as additional implicit regularisation when single-subject sample sizes are small (about 440 per subject in the training fold). For Phase 2 the recommended default is **pooled**, since it produces stabler regression numbers without sacrificing classification quality. See Figure 3 for the cell-level comparison.

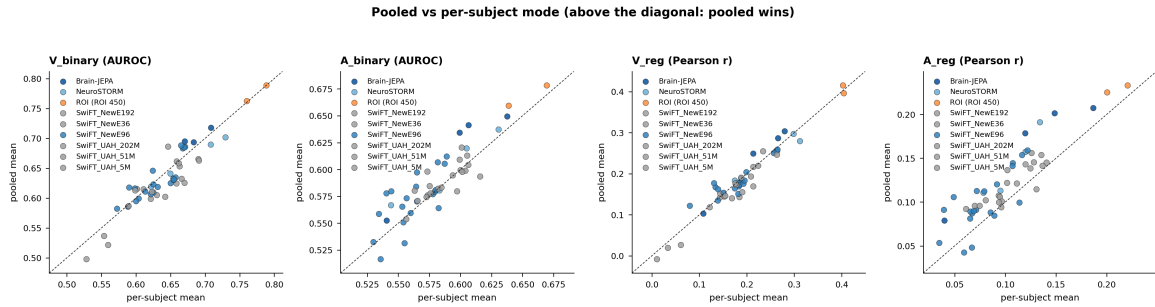


Figure 3: Pooled vs. per-subject mode for each (feature, init, padding, head) cell on each task. Points above the diagonal indicate the pooled mode is better. For regression tasks (rightmost two panels) the cloud sits modestly above the diagonal, consistent with the small positive Δ reported in the main text.

4 Discussion

4.1 What this benchmark establishes

This Phase 1 report establishes, under a single common protocol, three results.

1. The three brain foundation models tested in their frozen state do not yield embedding representations of naturalistic emotion fMRI that exceed a 450-region mean BOLD ridge baseline.
2. Pretrained video features set a high stimulus-prediction ceiling that brain features fall systematically short of, ranging from +0.13 to +0.18 AUROC on the binary tasks and +0.19 to +0.35 Pearson r on the regression tasks.
3. Within the SwiFT frozen probe, no padding strategy that preserves temporal information meaningfully outperforms a temporal-mean control, suggesting that the frozen SwiFT embedding on these short padded inputs is essentially a spatial code.

These three findings together delineate the gap that any trained joint architecture in Phase 2 must close. The gap is not small, and the contribution of brain information will need to come either from (a) joint training that exploits brain features the frozen probe cannot expose, or (b) reframing the evaluation toward subject-specific subjective response that the video features cannot, in principle, predict.

4.2 Why ROI beats BFM, in plain terms

The Schaefer-400 plus Tian-S3-50 mean BOLD vector is a fixed, low-dimensional summary of the exact same input that the BFMs see. It carries no learned temporal pattern, no learned spatial inductive bias, and no pretraining-derived prior; only a parcellation. That this representation outperforms three pretrained Transformer-based BFMs on every task should be read as evidence that, for stimulus-level emotion prediction under heavy temporal padding, the BFMs’ learned representations are not contributing positive task-relevant information beyond what raw mean BOLD already provides. The most plausible reason is the combination of pretraining-test domain mismatch and the 75% padding ratio that defines our evaluation set.

This is a frozen-probe statement, not a verdict on the BFMs. We want to be precise about what the present results do and do not show. The result is that the *frozen embeddings* of three resting-state-pretrained BFMs are dominated by a parcel-wise temporal mean. It is *not* a statement that these BFMs are uninformative for naturalistic emotion fMRI. Several mechanisms could in principle recover the gap that frozen probes leave on the table. Among them are end-to-end fine-tuning of the BFM on emotion targets, parameter-efficient adaptation (e.g., LoRA on a subset of layers), and trained joint architectures that combine the BFM output with a stimulus-side encoder under a shared loss. The present benchmark establishes that the resting-state self-supervised pretraining paradigm, when used as a fixed feature extractor under heavy temporal padding, does not by itself yield representations competitive with simple parcel-wise summaries. Whether that paradigm can be made competitive by trained downstream use is an empirical question that Phase 2 is designed to answer.

4.3 Implications for Phase 2

The headline take from this benchmark is that the conventional recipe (resting-state self-supervised pretraining as a feature extractor, frozen probe on downstream emotion targets) does not transfer

to naturalistic emotion fMRI under our protocol. A different recipe is required, and Phase 2 is positioned to test what that recipe should be. We identify two non-mutually-exclusive directions.

1. **Replace the frozen probe by trained integration.** The frozen-extraction regime discards the largest knob the BFMs have for adapting to a new task: their internal parameters. Trained joint architectures (LLM-token injection, cross-attention, contrastive alignment, late fusion) all provide a path for the BFM representation to be shaped by the emotion target. If any of these architectures improves over Tier 3 video features by even a small margin, that is the first quantitative demonstration on this benchmark that brain features carry information beyond what the stimulus alone provides. Conversely, if none does, the negative result is itself informative about the regime.
2. **Reframe what brain features are expected to add.** Pretrained video models predict the stimulus-level emotion label almost perfectly because that label is, by construction, a property of the stimulus. Subject-specific subjective experience (the same clip eliciting different affective states in different viewers) is not predictable from the video alone and is the natural locus of brain contribution. Two concrete operationalisations for Phase 2 and 3 are: (a) counterfactual subject swap, in which the same video is paired with a different subject’s brain response and the resulting model output is required to change systematically; (b) generative subject-conditioned captioning, in which the model produces an emotion-aware description that varies across subjects for the same input clip. These targets cannot be saturated by the video alone and thus expose a measurement axis on which brain features can in principle dominate.

Neither path is guaranteed to succeed, but the benchmark presented here makes precise what success would look like. It also raises the possibility, which we should be willing to entertain, that the gap to video features on stimulus-level targets cannot be closed with the present dataset size (five subjects, 2185 stimuli). Extension to larger naturalistic emotion fMRI corpora (e.g., Emo-Film, REELMO) is the obvious avenue if stimulus-level prediction remains the benchmark of interest.

4.4 Limitations

Five constraints bound the generalisability of the present results. First, the dataset comprises only five subjects, which limits statistical power for any between-subject claim. Second, only a single seed is currently run for MLP heads, so small differences between MLP cells should not be over-interpreted. Third, the `Cat34_top1` and `Dim14_multi` tasks were extracted but are not the focus of this report; they appear in the supplementary CSVs. Fourth, padding ablation is performed only on SwiFT NewE96; Brain-JEPA and NeuroSTORM padding ablations were deemed scope. Fifth, the BFM `scratch` variants share an identical architecture with their resting counterparts; they isolate pretraining contribution from architecture contribution but cannot probe other choices such as window size or temporal patch size.

4.5 Outlook

The benchmark presented here is a publishable artifact in its own right, suitable for a NeurIPS Datasets and Benchmarks-style submission or a workshop venue. As a Phase 1 outcome it equips the project with three concrete things going forward: a fixed protocol to compare every Phase 2 trained model against, an honest characterisation of what frozen BFM probes can and cannot do on naturalistic emotion fMRI, and a clean motivation for why Phase 2 trained integration must close a measurable gap to be informative. The SwiFT model size and architecture variant comparison (§3.6) further constrains Phase 2 model selection by showing that simple SwiFT

scaling does not improve frozen-probe emotion performance, so the variant chosen for fine-tuning should be selected for engineering convenience rather than expected size advantage.

A Files and reproducibility

All raw per-fold result CSVs are written to `results/phase1/`. Naming convention:

- Brain probes: `bfm_probe_<feature>[_<task>].csv`
- Video probes: `video_probe_<feature>.csv`
- SwiFT padding ablation: `swift_padding_ablation_<task>.csv` and `swift_padding_cyclic_only_<task>.csv`

Per-row aggregation: `code/probes/_summary_helper.py`. Per-task benchmark tables: `code/analysis/phase1_bench`. Figures: `code/analysis/phase1_unified_analysis.py`. Detailed extended tables appear in the accompanying *Supplementary Material*.

B SwiFT variants extraction details

The variants comparison reported in § 3.6 extracted five additional SwiFT checkpoints under **zero** padding (matching the main-grid default established in § 2.4). This appendix lists the architectural specifications of each variant and the extraction and probe commands used.

B.1 Variants included

- **UAH P1 5M** (ver9, $d_{\text{embed}} = 36$, patch [6, 6, 6, 2]).
- **UAH P2 51M** (ver9, $d_{\text{embed}} = 96$, patch [6, 6, 6, 2]).
- **UAH P3 202M** (ver9, $d_{\text{embed}} = 192$, patch [6, 6, 6, 2]).
- **NewE36 (NewUAH newE36)** (ver11, $d_{\text{embed}} = 36$, patch [6, 6, 6, 1]).
- **NewE192 (NewUAH newE192)** (ver11, $d_{\text{embed}} = 192$, patch [6, 6, 6, 1]).

NewE96 is already included as the main SwiFT entry above; UAH P3 806M and the HCP subject-scaling variants are excluded by scope.

B.2 Padding choice

Variants were extracted under **zero** padding only, matching the convention established in § 2.4 for all BFMs in the main grid. The padding ablation in Table 6 confirmed that the choice among **zero**, **spatial_only**, **mean**, and **cyclic_replicate** is within 0.001 of the overall mean, so this single padding choice is representative.

B.3 Extraction and probe commands

Extraction (5 variants $\times$ 2 init $\times$ 5 subjects = 50 cells, $\sim$ 6 hours wall clock).

```
bash /pscratch/sd/s/sjmoon/FEELIN/code/bfm_embeddings/\
run_full/extract_swift_variants_with_padding.sh zero
```

Output paths follow

`output/embeddings/swift_<TAG>_SL20_<init>_pad-zero/sub-XX.pt`

with `<TAG>` in {NewE36, NewE192, UAH_5M, UAH_51M, UAH_202M} and `<init>` in {resting, scratch}.

Probe (5 variants $\times$ 4 tasks = 20 runs, parallel across nodes). Per-variant per-task wrappers are at `code/probes/wrappers/bfm/Swift_<TAG>/<task>.sh`, with task-level master scripts at `code/probes/wrappers/bfm/_task_masters/<task>_all_swift.sh` that loop sequentially through all six Swift variants. The probe outputs merge into the per-task ranking by re-running `code/analysis/phase1_unified_analysis.sh` and `code/analysis/phase1_benchmark_table.sh`.

Note on Brain-JEPA and NeuroSTORM. No additional padding modes are planned for Brain-JEPA or NeuroSTORM in this phase. The Brain-JEPA per-ROI temporal patch already averages within a 16-TR window, and the Swift ablation showed that the choice among the four non-replicate paddings is below resolution. Extending the ablation to the other two BFMs is left as supplementary work.